5. Scheduling Program for Highest Priority

Code:

```c
#include <stdio.h>

int main()
{
    int n, i, j, temp;
    int bt[20], p[20], pr[20], wt[20], tat[20];
    float avgwt = 0, avgtat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        p[i] = i + 1;
        printf("Enter burst time and priority of P%d: ", p[i]);
        scanf("%d%d", &bt[i], &pr[i]);
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (pr[i] > pr[j])
            {
                temp = pr[i];
                pr[i] = pr[j];
                pr[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
```

```
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++)
        wt[i] = wt[i - 1] + bt[i - 1];

    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
        avgwt += wt[i];
        avgtat += tat[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for (i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
               p[i], bt[i], pr[i], wt[i], tat[i]);

    printf("\nAverage Waiting Time = %.2f", avgwt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avgtat / n);

    return 0;
}
```

Output:

```
Enter number of processes: 5
Enter burst time and priority of P1: 3
6
Enter burst time and priority of P2: 5
4
Enter burst time and priority of P3: 3
7
Enter burst time and priority of P4: 1
2
Enter burst time and priority of P5: 4
4

Process BT      Priority        WT      TAT
P4      1       2               0       1
P2      5       4               1       6
P5      4       4               6       10
P1      3       6               10      13
P3      3       7               13      16

Average Waiting Time = 6.00
Average Turnaround Time = 9.20
```